

Build a Foundation for AI with Integration

Unlock AI-powered composability with MuleSoft.



Contents

Executive Summary	03
Introduction	04
1 Part 1: API Development Fundamentals	05
APIs are building blocks	
Craft your API specification	
Implement your API vision	
2 Part 2: Build an Application Network	08
The concept of an application network	
Utilize Anypoint Exchange for universal connectivity	
3 Part 3: Extend the Application Network	10
AI-powered integration	
Data-powered integration	
Modernize B2B integrations via EDI and APIs	
Power integrations for industry-specific solutions	
Conclusion	16



Executive Summary

Integration is critical in consolidating and unlocking data across various business landscapes – meanwhile enterprise organizations have an average of [991 apps](#), and only [28% of these are connected](#). That's why [62% of IT leaders](#) feel that their organization is unprepared to fully harmonize their data systems to leverage AI technologies.

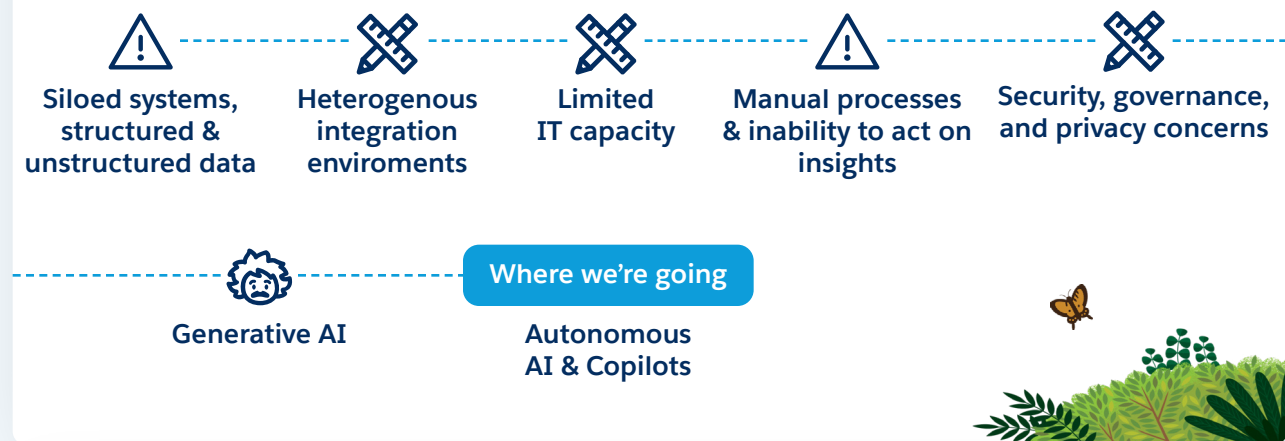
Customer expectations are higher than ever, with [39% of customers](#) reporting they have taken their business elsewhere when their expectations weren't met. To rise to the new level of demand, organizations must ensure their data is accessible across the enterprise to streamline operations and enable a competitive edge.

MuleSoft enables organizations to harness the power of APIs to automate processes and connect data systems. This in turns makes it easier to leverage AI, helping meet increasing customer expectations.

This whitepaper explores MuleSoft's integration product portfolio, emphasizing the necessity of integration to realize AI's full potential.

3

Ongoing Challenges Stand in Our Way



83% of IT Leaders say AI has increased developer productivity
Source: [2024 Connectivity Benchmark Report](#)

991 the number of average applications used by businesses
Source: [2024 Connectivity Benchmark Report](#)

Introduction

Organizations have experienced many disruptive technology shifts over the last two decades – from the adoption of SaaS, to migration to the cloud, and the rise of automation. While these changes might feel as if they have happened over a long time, many companies still struggle to keep up.

With the rapid rise of generative AI, IT teams are under increased pressure, as [86% of IT leaders expect](#) generative AI to play a big role in their digital transformation. Yet organizations today are already dealing with a complex web of siloed systems and data, heterogeneous metadata across environments, and a lack of security and governance.

To embrace new technologies and keep up with industry trends, an organization's entire collection of digital building blocks, including data, LLMs, and processes, must be streamlined and connected.

Because of Salesforce Einstein, AI has been woven into the fabric of MuleSoft's product ecosystem. This underlying AI platform empowers developers to effortlessly leverage advanced AI capabilities. The Einstein Trust Layer makes it possible to weave AI into applications and integrations in a secure and controlled way. And Einstein's chat-based experiences allow developers to build APIs and integrations from natural language prompts.

Salesforce Einstein integrates seamlessly across MuleSoft's offerings, enabling smarter and faster decision-making, automating routine tasks, and providing insights that drive more personalized and effective customer engagements. Through this integration and APIs, MuleSoft simplifies the complexities of digital transformation and equips developers with the tools to work faster, smarter, and more intuitively in today's AI-driven landscape.

To begin your integration journey, it is critical to understand the fundamentals of APIs.

80% of IT leaders report already using multiple AI models, and the mean number of AI models used by IT organizations is estimated to increase by 69% within the next three years. Gartner predicts more than 30% of the increase in demand for APIs will come from AI and tools using LLMs by 2026.

Part 1: API development fundamentals

APIs are the foundation of modern-day integration. They are the core elements of delivering business capabilities that balance innovation, speed and operational stability. A single API is a gateway to information or data from a specific system or application.

However, APIs' true value lies in their ability to be used together. When combined, APIs can craft robust digital experiences that seamlessly incorporate hundreds of back-end systems.

APIs are building blocks

Today, virtually every digital experience we interact with is powered by APIs including mobile apps, websites, smart home devices, and streaming services. This is why APIs are often thought of as the building blocks for composed applications.

By connecting them, you can quickly add functional components for your customers and employees in systems, such as:

- Checkout systems
- Shipping cost estimators
- New mobile experiences

The modularity of APIs is a critical benefit in today's digital-first world. This agility allows organizations to quickly add new capabilities to their existing experience and adjust previously built projects.

By leveraging your organization's APIs as building blocks, you can build vibrant experiences for your customers while maintaining agility and scalability. Furthermore, you can ensure your experiences can grow and evolve with the demands of the market and the ever-changing expectations of users.

As we look to the future, APIs are positioned to continue playing a vital role. With the rise of AI, streamlined access to data from various systems is more important than ever.

To harness the full potential of artificial intelligence, a loop between disparate systems and the growing sea of large language models (LLMs) is required. Properly integrated and managed APIs are the key to unlocking this potential. Adding LLMs to your digital estate without a clear integration strategy is a recipe for disaster.



Craft your API specification

An API specification (spec) is the blueprint for your API and outlines its functionalities, data formats, and endpoints. It's a critical step in API development and ensures clarity and alignment among stakeholders before coding begins.

Best practices suggest that your API spec accommodate both the RESTful API Modeling Language (RAML) and the OpenAPI Specification (OAS) formats.

MuleSoft offers extensive support for API spec development through the Design Center in Anypoint Platform, facilitating cloud-based, collaborative spec development, featuring a configuration-driven low-code experience. This allows users to efficiently outline API structures without deep coding knowledge, making the process accessible and streamlined.

Key features: Anypoint Platform Design Center

Automated documentation - Enable developers to generate API documentation automatically

Mocking service - Test APIs with sample or mock data

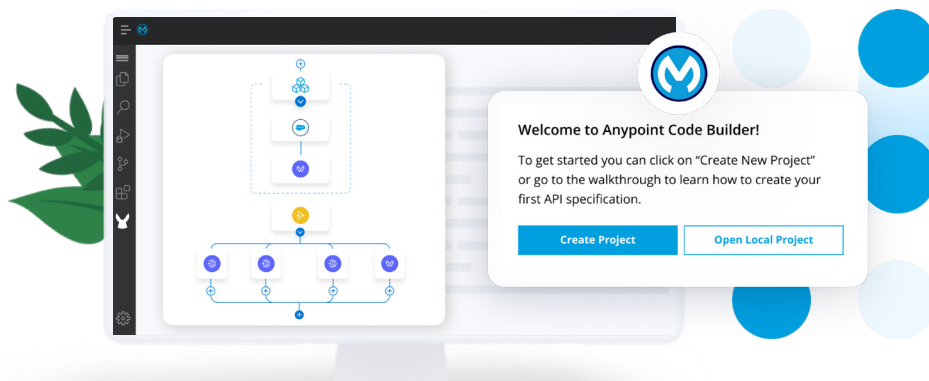
Share API models - Collaborate and adjust APIs at the design stage



MuleSoft product highlight

MuleSoft's Anypoint Code Builder is a next-gen cloud-based IDE that seamlessly integrates spec writing with API implementation. This ensures a fluid development journey from spec-to-code and offers a comprehensive environment for end-to-end API development. MuleSoft streamlines API spec creation through these tools, making it easier for teams to develop, test, and refine their APIs efficiently.

[Learn more](#)



Implement your API vision

After completing the API specification, teams must add business logic to our API outline. MuleSoft's integrated platform makes it easy to import the specification into an integrated development environment (IDE) – Anypoint Studio or the newer VS Code-based Anypoint Code Builder. The IDE scaffolds the API into a template flow for each HTTP method so that developers can focus on crafting the specific business logic for the API without worrying about configuration and setup.

Accelerate development with connectors and AI

MuleSoft's online marketplace, Anypoint Exchange, provides hundreds of connectors to common business systems, whether it's Snowflake, Amazon S3, NetSuite, or something else. This marketplace also hosts accelerators (industry-specific pre-packaged connectors and templates) and examples to speed up the API development process significantly.

Furthermore, Einstein for Anypoint Code Builder offers a natural language experience that can write code, offer suggestions, and automate tasks to streamline the creation of APIs and integrations. With the power of connectors and AI-assisted development, developers can build robust APIs and integrations quickly and efficiently.

Simplifying deployment, governance, and monitoring

The final step in bringing your API to life involves deployment, governance, and monitoring. With one-click deployment capabilities to CloudHub, our managed cloud service, deploying your API is a seamless experience. Anypoint Platform simplifies [API management](#) and security, offering intuitive interfaces to apply API policies and enforce [governance](#) standards. Additionally, MuleSoft provides robust monitoring tools to ensure your APIs are performing optimally and securely.



Part 2: Build an application network

The application network is a strategic framework of interconnected applications and services that facilitates seamless communication and data exchange through standardized integration methods.

Imagine a city road system where roads (APIs) connect various locations (applications, systems, services) efficiently and securely, ensuring smooth transit (data exchange) for all inhabitants (end users). The end goal is to organize the way for different applications, systems, and services to connect and communicate with each other seamlessly in a secure, efficient, and scalable manner.



Ultimately, the APIs you build are the only way to reach your end users. If you don't plan before you start building, you'll quickly end up with inefficiencies, redundancies, or even outages.

Your IT teams will waste time maintaining the integrations you built instead of creating new or reusing APIs to connect to your various systems and applications. The result? Your end users' experiences will suffer.

Utilize Anypoint Exchange for universal connectivity

At the core of MuleSoft's universal connectivity vision and services is Anypoint Exchange, a centralized hub for sharing and discovering integration assets. Developers can share their API specs, implementations, integrations, and deployed APIs with full documentation and access control across their company to promote a culture of reuse and collaboration.



Connectors

Connectors are pre-built integration tools that seamlessly connect popular applications, databases, and APIs.

Implementation templates

Implementation templates serve as blueprints for common integration patterns and scenarios. They provide a starting point for developers, offer best practices, and reduce the time it takes to deliver new integrations.

Accelerators

Accelerators are pre-packaged solutions designed to address specific business challenges or industry needs. They combine connectors, templates, and best practices to provide a comprehensive solution that can be quickly deployed and customized.

These three assets ease the burden on IT teams, allowing them to focus less on developing and managing integrations and empowering them to achieve universal connectivity and internal collaboration. Working together to build reusable assets is key to an organization's success.

Using building blocks to construct the composable enterprise makes it easy to adopt new technologies and swap out systems as needed without changing the entire technology stack.

MuleSoft product highlight

MuleSoft's [Catalog CLI](#) and [API Experience Hub](#) make it even easier to share assets both internally and externally.

[Anypoint API Sharing](#) makes it easier to share APIs as low-code connectors with your business users. This means developers and business users can collaborate on technology solutions, and IT can empower business users to build their automations using shared connectors. This extends the power of the unified platform to the business organization and eases the burden on IT and development teams.



Part 3: Extend the application network

AI-powered integration

AI is an unrelenting force that has reshaped how businesses approach integration and automation. MuleSoft stands at the forefront of this evolution and offers two sides to AI in integration: AI-powered tools to accelerate integration development and AI to enhance integration experiences through connected systems and better grounding.

4 Steps to AI success

1

Connect

Build APIs on a unified platform to connect and unlock **data** necessary for effective AI.

2

Compose

Reuse existing APIs and discover new APIs to compose new AI applications and **interfaces**.

3

Control

Manage all APIs including LLMs & AI **models** to protect sensitive information and control access & usage.

4

Action

Take action via API or RPA bot to drive **outcomes** across relevant systems and channels.

10

AI-powered tools for integration development

At the heart of MuleSoft's approach to AI-powered development is the integration of AI into traditional processes, making it easier for businesses to leverage AI when developing APIs and integrations. Working together means using a natural language experience, and leveraging the capabilities of generative AI to provide users with an intuitive starting point for creating integration flows.

Using natural language conversations, developers can swiftly generate the skeleton of their integration projects, significantly reducing the initial complexity and time investment. This AI-enhanced guidance streamlines the development process and democratizes access to advanced integration capabilities, enabling a more comprehensive range of professionals to contribute to integration projects.

AI-enabled experiences

MuleSoft also enables the direct use of AI in integrations and APIs to help create AI-driven customer experiences. Furthermore, our expansive library of connectors is a cornerstone of this acceleration, which plays a pivotal role in connecting disparate systems and data sources. This comprehensive collection ensures that business data can be seamlessly integrated into AI systems regardless of origin.

Think of a world where all your knowledge about your business and customers is stored in a single data lake with a unified metadata model. Your AI tools can leverage this data from this single source of truth to provide accurate and helpful responses. A support agent can make product recommendations or help troubleshoot a problem without needing to study the entire product library, or AI can automatically identify customers who are more likely to buy, given their browsing and purchase history.



Data-powered integration

MuleSoft's approach to data integration addresses the entire data lifecycle, from ingestion to processing. This holistic methodology consists of three steps and is built on out-of-the-box connectors.

01. Trigger flows on system changes

Keeping data flows up-to-date with real-time system changes is crucial for maintaining data accuracy and relevance. MuleSoft uses triggers that detect file and system modifications to initiate processes as soon as changes occur. This capability is enhanced by features such as watermarking and custom Regular Expressions (Regex), allowing for granular control over which data changes prompt integration actions. This ensures that data systems are always synchronized and that only the most pertinent data is processed, maintaining the integrity and currency of the integrated data.

02. Transform the data with DataWeave

Once data is ingested, the transformation stage begins. With DataWeave, organizations can map, transform, and normalize data across different formats and structures. The DataWeave Playground allows for exploration in a sandbox environment where developers can model and test their DataWeave scripts. This secure environment ensures the accuracy and efficiency of data transformations before they are deployed.

03. Bulk data processing: batch jobs

The final step is implementing batch jobs. These are designed to handle large volumes of data efficiently and are highly customizable. Batch jobs allow developers to adjust parameters such as batch size and aggregator size to optimize the processing workload. This flexibility ensures that large datasets can be processed in a time-efficient and resource-effective manner.

Through a comprehensive data integration strategy, businesses can manage their data integration needs from the initial connection to data sources to the sophisticated processing of large datasets. With this approach, businesses can leverage their data assets to their fullest potential to drive insights, decisions, and actions that propel them forward in the digital landscape and adapt to rapid industry changes like AI.



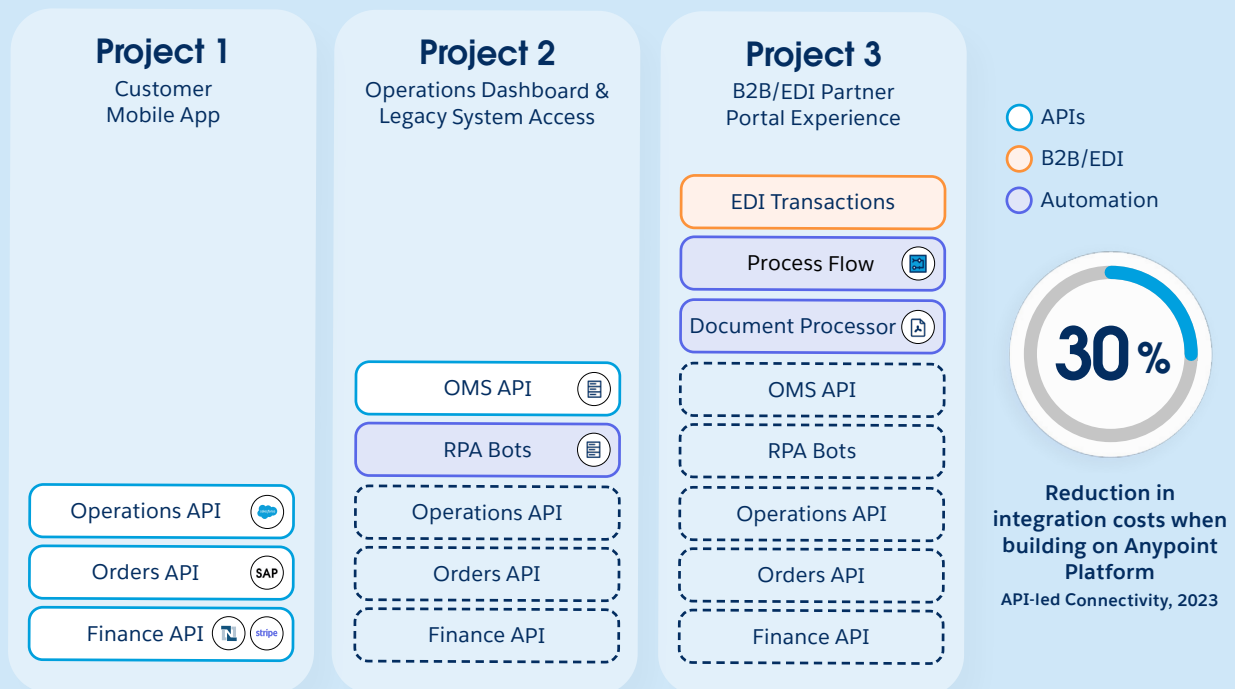
Modernize B2B integrations via EDI and APIs

In your average supply chain ecosystem, suppliers send and receive thousands of business transactions and communications with their networks of buyers daily to execute important processes like order placement, inventory confirmations, product fulfillment, and much more. Beyond order management, suppliers and buyers must communicate with their transportation and logistics partners to manage shipment and delivery processes for these orders.

This results in a lot of transaction data from order placement to delivery, all of which need to be structured and tracked. A composable application network within an organization makes it easy to add and adjust APIs that can ingest incoming orders or transactions from partners, process them with internal business logic, and route them to the correct back-end systems.

Traditionally, these business transactions are exchanged with partners via EDI (Electronic Data Interchange), the standard for B2B communication. The complexity of different transactions, data formats, and compliance regulations is difficult to manage without the format standards that EDI has provided.

AI-powered composability helps companies innovate faster



Although EDI will remain an important part of building a composable partner ecosystem, businesses are increasingly opting for modern communication methods such as APIs and document exchanges. This highlights the need for a platform to handle various types of B2B transactions using different communication methods and data structures.

MuleSoft Anypoint Partner Manager provides a low-code, configuration-driven experience that makes it easy to set up your business profile and communicate with partners across various formats without writing code. It is built with the complex needs of supply chain management in mind, with a simple interface that allows enterprises to onboard their partners faster, purpose-built tools that drive efficient API reuse, and point-and-click business transaction management.

With next-gen transaction monitoring and visibility, and transaction replay capabilities, MuleSoft enables teams to unlock important business and operational insights as you collaborate with your business partners.

Power integrations for industry-specific solutions

Industry-specific technology demands are increasing as organizations are pressed to adopt interoperability standards, such as FHIR for healthcare and TM Forum for communications. Anypoint Exchange provides many industry-specific assets:

Industry accelerators

Accelerators package together connectors, APIs, templates, and other technical assets for a given use case, and they are particularly valuable for organizations looking to implement complex integrations in a short timeframe. Imagine taking only days instead of months to connect your patient data in Salesforce Health Cloud to your healthcare information management systems, or to automate refund processing for your product returns.

To date, we have Accelerators for specific industries, including financial services, healthcare and life sciences, retail and consumer goods, and manufacturing. These accelerators include pre-built integrations to power top industry initiatives. We also have accelerators for key systems, including SAP and Commerce Cloud, Service Cloud, and Data Cloud.



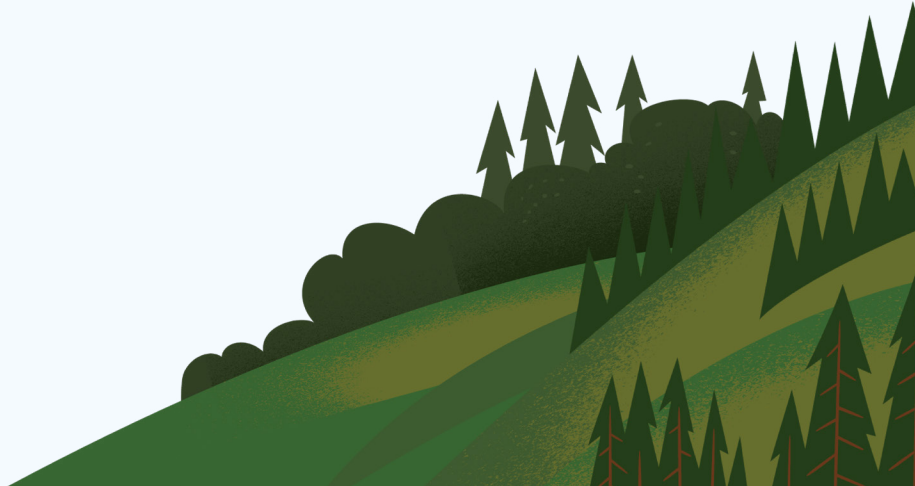


In the financial services sector, Accelerators can streamline the integration of banking systems with Salesforce, enhancing customer experiences and operational efficiency. And in healthcare, they facilitate the secure exchange of patient data between systems, improving patient care and compliance with regulations.

MuleSoft Direct

While accelerators can help jumpstart your integrations, MuleSoft Direct allows users to simply log in to Salesforce setup and connect with their MuleSoft Anypoint Platform account. This offers them common industry-specific integrations that they can enable using clicks, which can be customized on the back end of Anypoint Platform.

Today, MuleSoft Direct is available for financial services, healthcare, communications, energy and utilities, consumer goods, media, loyalty management, automotive, and Manufacturing Cloud. We plan to onboard additional Salesforce Clouds in the future.



Conclusion

MuleSoft's mission has always been to provide a robust suite of tools designed to streamline the integration of diverse systems, applications, and data across any cloud or on-premises environment. Today, MuleSoft's portfolio extends into low-code tooling for automation, comprehensive API management, and one-click deployment options for various cloud platforms.

With customer expectations higher than ever, organizations must adopt an agile and efficient IT estate to meet these expectations. And thanks to AI's widespread adoption, things only get more complicated. By establishing a clear integration strategy, organizations can use APIs to create experiences that will delight customers.

With MuleSoft's integration solutions, organizations can easily connect systems and data to create a connected and agile enterprise equipped with composable building blocks. This composability is thanks to the power of APIs, which enable businesses to easily navigate the challenges of digital transformation and pave the way for innovation, efficiency, and growth.

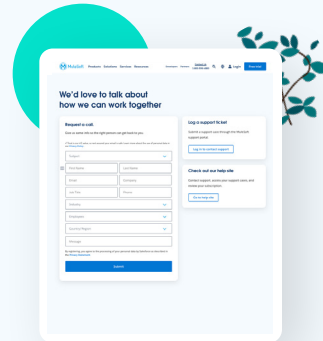


What's next?

Adopt a trusted platform for automation and more.

Learn how MuleSoft can help your team increase productivity, democratize innovation, and ultimately create exceptional customer experiences.

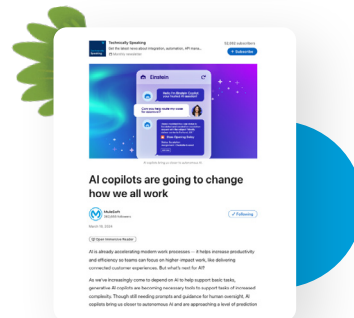
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See the next generation of integration development: Anypoint Code Builder.

Discover how MuleSoft's AI-enabled, cloud-based IDE allows developers to design, develop, and, deploy APIs, integrations, and automations – all from a single environment.

[Watch video](#)



Secure your digital estate with these best practices.

Guide your organization to develop a comprehensive API security strategy, and demonstrate why your strategy must be developed as soon as possible.

[Get the whitepaper](#)





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